Network orientation and cross-calibration

SLANT network study. Computed 8 October 2026 for 100 host orbits over one year.

Every unit in the baseline network faces the wake and records sunsets, so the network samples the evening terminator. This note tests an alternative for large networks: units on the higher hosts face forward along the orbit (the ram) and record sunrises, so the network samples dawn as well. It compares coverage and the in-flight cross-calibration of the two networks.

The altitude threshold, chosen before any network result

A dawn-minus-dusk difference measured by the network carries the uncertainty of the offset between the ram group and the wake group. For independent unit offsets that uncertainty scales as √(1/N_ram + 1/N_wake), which is smallest when the two groups are equal. Putting the ram half on the highest hosts minimises the atomic-oxygen dose on the ram-facing filters. The threshold is therefore the median host altitude of the population, 567.1 km, rounded to **565 km**.

Host altitude	One-year ram atomic-oxygen fluence, atoms/cm² (NRLMSIS 2.1 global mean, ap 4)		
	F10.7 = 70	F10.7 = 140	F10.7 = 200
400 km (lowest host)	5.1×10^{20}	2.3×10^{21}	4.6×10^{21}
500 km	5.9×10^{19}	4.5×10^{20}	1.1×10^{21}
565 km (threshold)	1.5×10^{19}	1.6×10^{20}	4.8×10^{20}
650 km	2.8×10^{18}	4.5×10^{19}	1.6×10^{20}

Ram speed is the orbital speed less a mean corotation term (7.26 km/s at 565 km); the figures are approximate to that level. Whether the Zr, Al and In films tolerate 4.8×10^{20} atoms/cm² in a year has not been tested.

Ram-facing units: 1 of the first 6 hosts, 13 of 30, 54 of 100. Nine of the 54 are sun-synchronous hosts in near dawn-dusk planes that record no occultation in either orientation, so 45 ram-facing units and 46 wake-facing units produce data.

Coverage

Both configurations give 10.33 occultations per unit per day and 70 % of unit-days with at least one, as the symmetry between sunrise and sunset requires. In the mixed network the ram-facing units average 9.2 per day (62.8 % of unit-days) and the wake-facing units 11.7 (78.6 %), because the nine silent hosts are all in the ram group.

Local solar time of tangent points, N = 100	0–6 h	6–9 h	9–12 h	12–15 h	15–18 h	18–21 h	21–24 h
Wake-only	7.3 %	2.1 %	7.8 %	3.3 %	39.3 %	34.2 %	5.8 %

Local solar time of tangent points, N = 100	0–6 h	6–9 h	9–12 h	12–15 h	15–18 h	18–21 h	21–24 h
Mixed, all units	17.1 %	14.3 %	2.1 %	6.3 %	27.4 %	27.7 %	5.2 %
Mixed, ram-facing units	33.2 %	29.3 %	3.1 %	12.7 %	4.4 %	11.1 %	6.2 %

The share of events before 09 h rises from 9.4 % to 31.4 %. The wake-only network already records some dawn events: a few sun-synchronous hosts have orbital sunsets whose tangent points lie near 05 h local time.

Cross-calibration

Coincidences at 500 km and 30 min, per year:

Configuration	N	Pairs	Wake–wake	Ram–ram	Ram–wake
Wake-only	6 / 30 / 100	1,581 / 48,501 / 357,161	all	–	–
Mixed	6	760	760	0	0
Mixed	30	31,472	22,538	7,943	991
Mixed	100	226,963	122,276	70,851	33,836

The ram and wake groups share pairs because some ram-facing units on sun-synchronous hosts record events in the afternoon and evening. The wake-only total reproduces the 0.352 million pairs of the occultation-rate study.

Connectivity. Units that record no occultation are excluded from both. Wake-only: all units connected at 6 hosts. At 30 hosts, 27 of 29 active units are connected in every tolerance cell. At 100 hosts, 90 of 91 are connected. The units outside are sun-synchronous hosts (units 6 and 11) whose sunsets fall at dawn local time, where no other wake-facing unit observes. Mixed: 91 of 91 connected at 100 hosts in every cell, and all 29 at 30 hosts from 300 km / 30 min outward. At 6 hosts the single ram-facing unit has no partner.

The adjustment. The operating point is the coincidence tolerance that minimises the total per-unit error, with a 5 % allowance for atmospheric waves; N counts active units. The single-unit error combines the per-occultation error (0.75 % quiet, 1.49 % storm) with the offset left after adjustment, in quadrature.

Configuration, N	Operating point	Offset left, quiet / storm	Single-unit error, quiet / storm	Slowest mode, days	Condition number
No adjustment	–	0.84 / 2.25 %	1.13 / 2.70 %	–	–
Wake-only, 6	1000 km / 60 min, 750 km / 60 min	0.29 / 0.72 %	0.80 / 1.66 %	40 / 33	5.5 / 4.7
Wake-only, 30	500 km / 30 min, 200 km / 15 min	0.38 / 0.96 %	0.84 / 1.77 %	99 / 114	67 / 150
Wake-only, 89	300 km / 30 min, 200 km / 10 min	0.33 / 0.73 %	0.82 / 1.66 %	65 / 58	29 / 49
Mixed, 6	not connected	–	–	–	–

Configuration, N	Operating point	Offset left, quiet / storm	Single-unit error, quiet / storm	Slowest mode, days	Condition number
Mixed, 30	300 km / 30 min	1.15 / 4.93 %	1.37 / 5.15 %	568 / 320	1,266
Mixed, 91	300 km / 30 min, 200 km / 15 min	0.36 / 0.97 %	0.83 / 1.78 %	92 / 72	57 / 62

The wake-only network improves the single unit from 1.13 to 0.82 % quiet and from 2.70 to 1.66 % in a storm, and converges in 58 to 65 days at full size. The Kalman simulation agrees with the analytic random term to within 7 % for wake-only and 15 % for mixed.

Mixed at 30 units is worse than no adjustment. Its 991 cross pairs a year carry a systematic air-mass difference. At 300 km / 30 min that difference is 1.15 % against a sign-flip null of 0.53 % quiet, and 6.0 % against 3.6 % in a storm. The adjustment maps it into the offsets, and it is larger than the 0.84 % and 2.25 % offsets it is meant to remove. The condition number is 1,266. At 91 units the cross pairs number 33,836 a year, and the mixed network comes within 0.02 % (quiet) and 0.12 % (storm) of wake-only.

What follows

1. A mixed network adds dawn coverage (31 % of events before 09 h, against 9 %) at no cost in occultation count, and at full size its cross-calibration matches wake-only to within 0.02 % quiet and 0.12 % in a storm.
2. Below roughly 30 units a mixed network should not be cross-calibrated as one network: the ram group has too few partners, and its ram–wake pairs carry a systematic difference larger than the offsets. Dawn coverage is therefore a property of a large network, with ram-facing units on hosts above 565 km.
3. Ram-facing filters must survive at least 4.8×10^{20} atoms/cm² a year at 565 km, a fluence not yet tested.
4. The wake-only adjustment closes for 89 of 100 hosts.

The host population spans 400 to 832 km, so hosts between 350 and 400 km are not represented, and mounting error is taken as zero.

Code: `channel-plan/net_fov_mixed.py` and `channel-plan/run_net_adjust_fov.py`, in the code archive.